Qwen inference: local vs cloud

Decision brief for Qwen3.5-9B and Qwen3-235B-A22B | Inference only | USD

Practical starting point. For an internal pilot, Qwen3.5-9B at 4-bit on one 16 GB GPU is the simplest option. For the 235B model at 4-bit, plan around 192 GB of combined GPU memory and validate speed on the exact multi-GPU system before committing to a business SLA.

Hardware, speed and rental cost

Model / format	Weight file	Local compute target	Latency reference	Cloud GPU rent	730 h / month
Qwen3.5-9B 4-bit GGUF	6.17 GB [2]	1 x 12-16 GB GPU 16-32 GB RAM 25 GB free SSD	137 output tok/s on RTX 4090 [6]; ~1.9 s for 256 output tokens, plus prompt time	1 x RTX 4090 **$0.89/h**	**$650**
Qwen3.5-9B BF16	18.41 GB [2]	1 x 24 GB GPU 32 GB RAM 50 GB free SSD	No matching published BF16 measurement for the proposed rental GPU	1 x RTX 4090 **$0.89/h**	**$650**
Qwen3-235B-A22B 4-bit GGUF	142 GB [4]	~192 GB total VRAM 128-256 GB RAM 300 GB free SSD	14.7 output tok/s on 4 x H20 with a related INT4 format [5]; ~17 s / 256 tokens	2 x RTX Pro 6000 **$4.98/h**	**$3,635**
Qwen3-235B-A22B FP8	~235-260 GB*	~320 GB total VRAM 128-256 GB RAM 500 GB free SSD	71.7 output tok/s on 4 x H20 [5]; ~3.6 s / 256 tokens	4 x H100 PCIe **$11.56/h**	**$8,439**
Qwen3-235B-A22B BF16	~470 GB*	~640 GB total VRAM 256 GB RAM 1 TB free SSD	74.5 output tok/s on 8 x H20 [5]; ~3.4 s / 256 tokens	8 x H100 PCIe **$23.12/h**	**$16,878**

* FP8 and BF16 weight sizes are approximate from 235B parameters at 1 and 2 bytes per parameter. Runtime memory, context cache and vision inputs add overhead. Rental rates are listed GPU prices multiplied by GPU count [7]; multi-GPU availability and co-location must be confirmed. Monthly totals assume 730 billable hours and exclude storage, traffic, tax and support.

Speed figures use different hardware, quantizations and software from the rental examples. They describe generated tokens after prompt processing, not guaranteed request latency. The 9B observation used a short 512-token prompt and a different 4-bit quantization [6]; Qwen's 235B tests used 2,048 generated tokens with a 1-token input [5].

What business use adds

The GPU is only one part of a production service. Set these requirements before sizing capacity or promising response times.

Requirement	Why it matters	Minimum business decision / action
Use case and quality	The larger model may improve difficult tasks, but that benefit must justify its much higher cost.	Create a representative evaluation set: accuracy, hallucinations, tool use, language, and failure cases. Compare 9B and 235B on the same tasks.
Users and latency SLO	A single-user token rate does not predict performance at peak traffic.	Record peak concurrent users, prompt length, output length, requests/minute, and target p50/p95 time-to-first-token and full-response latency.
Availability	One GPU host is one failure point; large models also take longer to download and load.	Set uptime and recovery targets. Add a second replica, load balancing, health checks, and a fallback model when downtime is costly.
Data handling	Business prompts may contain customer, employee or confidential data.	Decide hosting region, retention period, encryption, access control, and whether prompts may enter logs. Review provider terms for sensitive data.
Security and abuse controls	An exposed inference endpoint can leak data or run up GPU costs.	Use authentication, TLS, network restrictions, tenant separation, rate limits, secret management, and vulnerability updates.
Monitoring and support	Production failures often appear as slow responses, OOM errors or low-quality outputs.	Track GPU memory, queue time, token rates, error rates, p95 latency, spend, and quality drift. Assign an owner and incident process.
License and governance	The model cards identify Apache-2.0, but your shipped stack may contain other components.	Review model and software licenses, preserve required notices, document model version and changes, and check output use rules for your product. [1][3]
Integration and modality	Application work may exceed model hosting effort.	Specify API, user interface, authentication, retrieval/search, tools and audit trail. Qwen3.5-9B accepts image-text input [1]; this Qwen3-235B variant is text generation [3].
Cost controls	Idle dedicated GPUs still accrue charges; duplicated capacity increases the bill.	Set spend alerts, retention limits, scale-down rules and a budget for storage, network, engineering and on-call support.

Capacity gate: Load-test your real prompt and response mix at expected concurrency. Size to the measured p95 target with memory headroom. Add replicas for resilience, then recalculate the total cost.

Cost and deployment decisions

These examples use public Runpod Pod rates checked 21 Oct 2026 [7]. They are planning arithmetic, not quotes or availability guarantees.

Example deployment	GPU rent / hour	40 h / month	730 h / month	With 2 replicas, 730 h / month
9B 4-bit or BF16 / 1 x RTX 4090	$0.89	$36	$650	$1,299
235B 4-bit / 2 x RTX Pro 6000	$4.98	$199	$3,635	$7,271
235B FP8 / 4 x H100 PCIe	$11.56	$462	$8,439	$16,878
235B BF16 / 8 x H100 PCIe	$23.12	$925	$16,878	$33,755

Replica figures assume two independent instances running continuously; actual high availability may also need a load balancer and data replication. Add persistent model storage, data transfer, taxes, logging, backups, security tools, engineering time and support. For on-premises hardware, add purchase/lease, power, cooling, networking and maintenance.

Recommended decision path

1. Define workload: users, peak concurrency, requests/day, prompt and output lengths, image use, data sensitivity and p95 latency target.

2. Prove value: test 9B first on a representative task set; use 235B only where its measurable quality gain warrants the cost.

3. Pilot the exact stack: chosen checkpoint, quantization, inference software and GPU layout; measure memory, first-token time, full response time and throughput.

4. Productionize: add access controls, monitoring, budgets, fallback, backups, incident ownership and a second replica if uptime requires it.

Sources and measurement notes

[1] Qwen3.5-9B model card (9B, image-text, Apache-2.0): https://huggingface.co/Qwen/Qwen3.5-9B

[2] Qwen3.5-9B GGUF sizes: https://huggingface.co/bartowski/Qwen_Qwen3.5-9B-GGUF

[3] Qwen3-235B-A22B model card (MoE, Apache-2.0): https://huggingface.co/Qwen/Qwen3-235B-A22B

[4] Official 235B GGUF sizes: https://huggingface.co/Qwen/Qwen3-235B-A22B-GGUF

[5] Qwen benchmark (4/8 x H20 96 GB; batch 1): https://qwen.readthedocs.io/en/latest/getting_started/speed_benchmark.html

[6] 9B RTX 4090 measured run (UD-Q4_K_XL, 512 context): https://llamabench.ai/?hardware=rtx4090&mode=tune&model=qwen3-5-9b

[7] Runpod GPU Pod prices (updated 8 Oct 2026): https://www.runpod.io/pricing